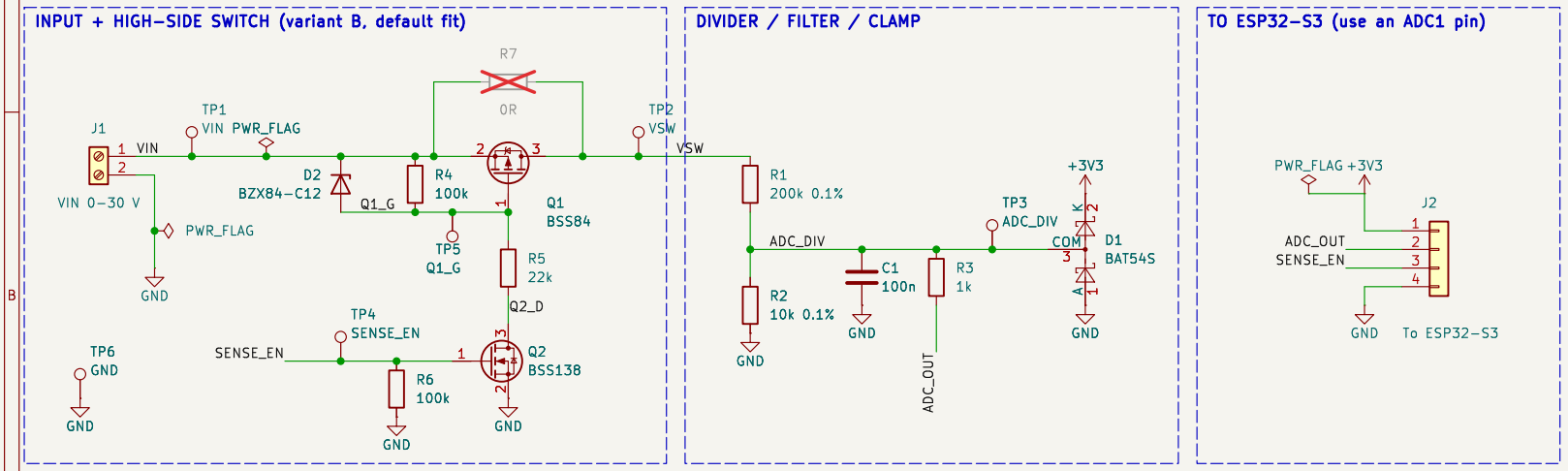
WARNING: Reference design. Test first with a current-limited supply and protection. Not for unattended use.

Design only, not fabricated, not bench-tested.



KEY VALUES (see README for derivations and sources)

Divider k = R2/(R1+R2) = 10k/210k = 1/21. Vin = 21 x V(ADC_DIV). 30 V -> 1.4286 V.

Use ESP32-S3 ADC1, ATTEN2 (6 dB): calibrated range 0-1600 mV, total error +/-10 mV (= +/-0.21 V at Vin).

Measurable up to 33.6 V (1600 mV x 21). Do not use ADC2 (shared with Wi-Fi).

Rth = R1||R2 = 9.52k. C1 = 100 nF (Espressif suggestion). tau = 0.95 ms, fc = 167 Hz, 12-bit settle 8.6 ms.

Worst-case ratio error: 1% parts +/-1.9% (+/-2.8% with 100 ppm/K, 0-70 C); 0.1% parts +/-0.19% (+/-0.41%).

D1 BAT54S clamps ADC_DIV to -VF..3V3+VF. VF max 240 mV @ 0.1 mA, 320 mV @ 1 mA. ESP32-S3 VIH max = VDD+0.3 V.

Worst fault (R2 open): clamp current = (Vin - ~3.55 V)/200k = 0.13 mA at 30 V. It flows INTO the 3V3 rail.

Variant B: drive SENSE_EN high, wait >= 12 ms (ngspice: 10.5 ms to 0.2 mV at 30 V), sample, drive low. Off drain <= 0.2 uA (datasheet max, 25 C).

Q1 |VGS| = 0.82 x Vin, clamped by D2 to 11.4-12.7 V (BSS84 VGS max 20 V). Switch needs Vin >= ~3 V.

Variant A (always-on): fit R7 (0R), omit Q1, Q2, D2, R4, R5, R6. Drain = Vin/210k (143 uA at 30 V).

Max VIN: 30 V operating. Q1 VDS max 50 V, R1 0805 working voltage 150 V.

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Abdulazeez Bello

Sheet: /
File: adc-input-scaler.kicad_sch

Title: ADC input scaler: 0-30 V to ESP32-S3 ADC

Size: A4	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. 9.0.3		Id: 1/1